

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To recommend an item to the user using collaborative-based filtering approach	
Experiment No: 12	Date:	Enrolment No:92301733012

Aim: To recommend an item to the user using collaborative-based filtering approach

IDE: Google Colab

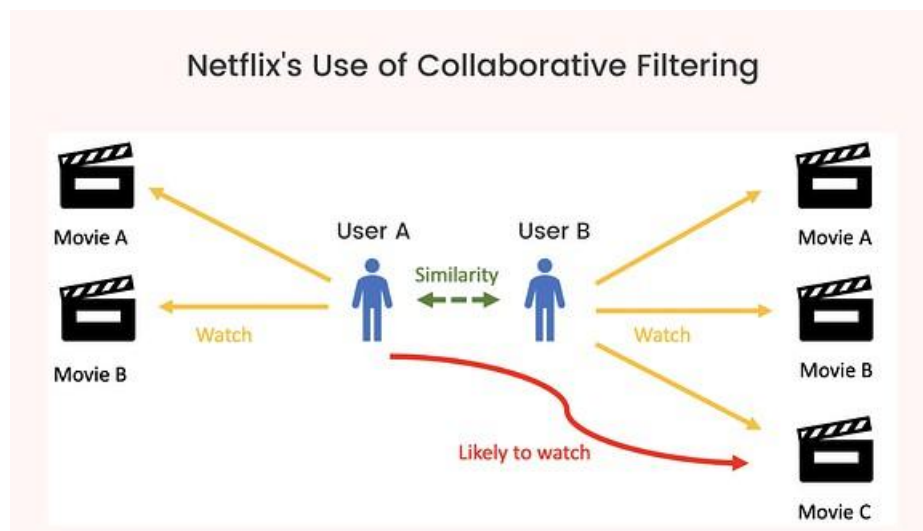
Theory:

There are a lot of applications where websites collect data from their users and use that data to predict the likes and dislikes of their users. This allows them to recommend the content that they like. Recommender systems are a way of suggesting or similar items and ideas to a user's specific way of thinking.

Recommender System is different types:

- **Collaborative Filtering:** Collaborative Filtering recommends items based on similarity measures between users and/or items. The basic assumption behind the algorithm is that users with similar interests have common preferences.
- **Content-Based Recommendation:** It is supervised machine learning used to induce a classifier to discriminate between interesting and uninteresting items for the user.

In Collaborative Filtering, we tend to find similar users and recommend what similar users like. In this type of recommendation system, we don't use the features of the item to recommend it, rather we classify the users into clusters of similar types and recommend each user according to the preference of its cluster.



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Pre Lab Exercise:

1. When can you use collaborative-based recommendation system approach?

Collaborative filtering is used when sufficient user interaction data is available and recommendations are based on similarities between users or items.

2. Write advantages of collaborative -based filtering approach

Collaborative filtering recommends diverse items based on user behavior without needing item features and improves as more data becomes available.

3. Write disadvantages of collaborative -based filtering approach

Collaborative filtering suffers from cold start, data sparsity, and requires large user data for accurate recommendations.

Program (Code):

```
!pip install pandas scikit-learn

import pandas as pd

from sklearn.metrics.pairwise import cosine_similarity

# Sample user-item rating matrix

data = {

    'Movie A': [5, 4, 0, 1],

    'Movie B': [4, 0, 0, 1],

    'Movie C': [1, 1, 0, 5],

    'Movie D': [0, 0, 5, 4],

}

df = pd.DataFrame(data, index=['User1', 'User2', 'User3', 'User4'])
```

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Fill missing values with 0

```
df = df.fillna(0)
```

Compute similarity between users

```
user_similarity = cosine_similarity(df)
```

```
user_sim_df = pd.DataFrame(user_similarity, index=df.index, columns=df.index)
```

Recommendation function

```
def recommend(user):
```

```
    similar_users = user_sim_df[user].sort_values(ascending=False)[1:]
```

```
    top_user = similar_users.index[0]
```

Recommend items liked by similar user but not by current user

```
user_ratings = df.loc[user]
```

```
top_user_ratings = df.loc[top_user]
```

```
recommendations = top_user_ratings[(top_user_ratings > 3) & (user_ratings == 0)]
```

```
return recommendations.index.tolist()
```

Test

```
print("Recommendations for User3:")
```

```
print(recommend('User3'))
```

Results:

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<pre> ... Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2) Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1) Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2) Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0) Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2024.1) Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2024.1) Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.14.1) Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.4.2) Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.5.0) Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil) (1.17.0) Recommendations for User3: ['Movie C'] </pre>		

Observation:

The system uses cosine similarity to find similar users and recommends items based on their preferences. It works well but depends on user data and suffers from sparsity and cold start issues.

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Post Lab Exercise:

1. Are you satisfied with the results/ratings shown by your designed model? If yes, justify.
If no, describe the probable reason for under performance of the system.

Yes, partially satisfied. The model provides reasonable recommendations based on user similarity, but accuracy can be improved with more data and advanced techniques.

2. Recommend top-10 “movies” using MovieLens 100K dataset
(Link: <https://grouplens.org/datasets/movielens/100k/>). Attach the code with output.
Also, write your observation for the same.

Code

```
import pandas as pd
from sklearn.metrics.pairwise import cosine_similarity

# Load data
ratings = pd.read_csv('ml-100k/u.data', sep='\t',
                      names=['user_id', 'item_id', 'rating', 'timestamp'])

movies = pd.read_csv('ml-100k/u.item', sep='|', encoding='latin-1',
                    names=['item_id', 'title'] + list(range(22)))

movies = movies[['item_id', 'title']]

# Merge
data = pd.merge(ratings, movies, on='item_id')

# Create matrix
user_movie_matrix = data.pivot_table(index='user_id', columns='title', values='rating').fillna(0)
```

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Similarity

```
movie_similarity = cosine_similarity(user_movie_matrix.T)
```

```
movie_sim_df = pd.DataFrame(movie_similarity,
                             index=user_movie_matrix.columns,
                             columns=user_movie_matrix.columns)
```

Recommend

```
def recommend(movie_name, top_n=10):
    return movie_sim_df[movie_name].sort_values(ascending=False)[1:top_n+1]
```

```
print("Top 10 Movies:\n")
print(recommend('Star Wars (1977)').index.tolist())
```

```
Top 10 Movies:
['Empire Strikes Back, The (1980)',
 'Return of the Jedi (1983)',
 'Raiders of the Lost Ark (1981)',
 'Back to the Future (1985)',
 'Indiana Jones and the Last Crusade (1989)',
 'Jurassic Park (1993)',
 'Terminator 2: Judgment Day (1991)',
 'Aliens (1986)',
 'Die Hard (1988)',
 'Fugitive, The (1993)']
```